BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Integrated Development Environments

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What does a compiler normally do?

2. Merge sorted [2, 7] and [4, 6]. Which result is ascending?

A) [2, 4, 6, 7]
B) [2, 7, 4, 6]
C) [4, 6, 2, 7]
D) [7, 6, 4, 2]

3. "A constant is a variable whose value must change after every instruction." Correct this. Explain why.

4. A bill is quantity x unit price + fee. "Calculations always run left to right." Correct this. How does operator precedence affect the result?

Today's objectives

Identify the four main IDE facilities named in J277. Explain what each does for a programmer.

Key vocabulary - Read, cover, recall

IDE
Integrated Development Environment: software bringing programming tools together.

Editor
Tool used to write and modify source code.

Error diagnostic
A message that helps identify an error and often its location.

Run-time environment
Facility for executing and testing a program.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Editor	The editor is where programmers write and change source code. Syntax highlighting and suggestions may make code easier to read and reduce typing mistakes.
Error diagnostics	Diagnostics report issues in code, such as missing brackets or incorrect syntax. A programmer can use the message and line number to find and correct the error.
Run-time environment	The IDE provides a way to run and test the program and inspect the output. This helps programmers check whether the program behaves as expected.
Translators	A compiler or interpreter provided within an IDE converts source code into executable instructions. Together, editor, diagnostics, translator and running tools support the development cycle.

I Do - One worked example

A pupil uses an IDE to create and test a quiz. Describe FOUR different IDE facilities they could use. [4 marks]

1. The editor lets the pupil enter and change the quiz source code [1].

2. Error diagnostics report code errors, helping the pupil find where corrections are needed [1].

3. A translator converts source code into instructions that can be executed [1].

4. The run-time environment runs the quiz so the pupil can inspect results and test the program [1].

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Explain how an editor's syntax highlighting and auto-completion can help a programmer. [4 marks]

2. Describe the purpose of the run-time environment and translator provided by an IDE. [4 marks]

3. An IDE reports 'missing closing bracket' at line 2. Explain how diagnostics and the editor help fix the problem. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Remove the unintended extra update so the program prints 7.

```
total = 5
total = total + 2
total = total + 2
print(total)
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. A learner writes and improves a quiz application. Explain how an IDE's editor and run-time environment help during development. [4 marks]

Q3. A programmer cannot run their new source code because it contains a syntax error. Explain how TWO IDE facilities help them get the program running. [4 marks]

Q4. A program should display 10 but displays 12. Describe TWO actions the programmer could perform using their IDE to investigate and fix this logic error. [4 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. An IDE's run-time environment helps a programmer execute and test code. ☐ T ☐ F
2. The editor is where programmers write and change source code ☐ T ☐ F
3. Diagnostics report issues in code, such as missing brackets or incorrect syntax ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 A compiler translates code before it runs. The executable can run again without translating unchanged source code. ☐

Q2 A - [2, 4, 6, 7] ☐

Q3 A named value that stays the same while the program runs. ☐

Q4 Multiplication comes before addition, giving 10 in the original example. Operator precedence affects the result. Use brackets to make the intended order clear. ☐

Code First - Model

```
total = 5
total = total + 2
print(total)
```

Output: 7

We Do - Check the model

1. Syntax highlighting makes different code elements easier to distinguish [1], making code easier to read [1]. Auto-completion suggests names or commands [1], which can reduce typing errors [1]. ☐

2. The run-time environment allows a program to be executed and tested [1], letting the programmer inspect behaviour and output [1]. A translator converts source code into a form the CPU can execute [1], enabling the program to run [1]. ☐

3. Diagnostics identify the possible issue [1] and help locate the line [1]. The editor lets the programmer change the code [1], after which they can run or test it again to check the correction [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 The editor lets the learner write and change source code [1], so they can improve the quiz instructions [1]. The run-time environment lets them execute the application [1], so they can check that the questions and scores behave correctly [1]. ☐

Q3 Error diagnostics identify the syntax problem [1], helping the programmer locate and correct it [1]. The IDE's translator converts corrected source into an executable form [1], allowing it to run [1]. ☐

Q4 Run the code and inspect the output [1] to establish where it differs from the expected value [1]. Step through the calculations or examine intermediate variable values [1], then correct the faulty line and run again [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: